

**BINAKAYAN RISING:AN ISOMETRIC DESKTOPROLE-PLAYING
GAME BASED ON THE HISTORICALBATTLE OF BINAKAYAN-
DALAHICAN**

**A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College -General Santos Inc.**

**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

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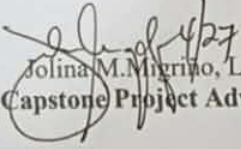
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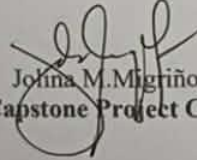
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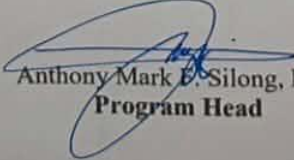

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INTRODUCTION

Project Context

Globally, the integration of digital technology into education has transformed traditional learning paradigms. The international rise of Game-Based Learning (GBL) and "serious games" has proven highly effective in fostering active engagement, critical thinking, and a deeper comprehension of complex subjects (Plass, Mayer, & Homer, 2020). International developers have successfully utilized interactive digital environments to preserve cultural heritage and simulate historical events, shifting learners from passive reading to active, spatial participation.

Transitioning to the national setting, the Philippine educational landscape is increasingly recognizing the value of digital integration. However, while the local game development industry is experiencing rapid growth, the creation of locally developed educational games that specifically tackle Philippine history remains largely underexplored. Recent pedagogical studies in the country confirm that gamified elements enhance motivation and learning gains. Specifically, researchers have demonstrated that localized, immersive role-playing games are highly effective in teaching Philippine revolutionary history by deepening students' cultural awareness and critical thinking (Moscoso et al., 2020).

Despite these national advancements, a critical gap exists at the specific, local historical level. A striking example of rich but underrepresented local history is the Battle of Binakayan-Dalahican (November 9-11, 1896) in Cavite, which marks the first major Filipino victory against Spanish colonial forces. This simultaneous two-front conflict showcased advanced military engineering—specifically the extensive trench networks designed by General Edilberto Evangelista—that successfully repelled naval bombardments and infantry assaults (Agoncillo, 1990). Unfortunately, current educational materials often reduce this dynamic, tactical battle to static textbook paragraphs, overlooking the ingenuity of the Katipuneros. To address this problem, there is a distinct

need to develop an interactive platform that directly translates the factual historical content, the geography, and the engineering of the Philippine Revolution into a modern, engaging medium.

Purpose and Description

The primary purpose of this project is to design and implement Binakayan Rising, an educational tactical role-playing game that serves as an interactive vessel for the historical content of the Cavite Campaign. By situating these historical facts within an asynchronous tactical framework, the project transforms static history into an experiential learning tool.

The proposed software is driven by its educational content through the following core components:

Deployment-Based Auto-Battler Combat: Integrates an asynchronous combat phase where player must strategically analyze the historical terrain(such as coastal shallows and Evanglista's fortifications) during a pre-battle deployment phase. Once units are placed, the AI executes the battle, mirroring the autonomous nature of real historical skirmishes.

Historically-Driven Recruitment System: Utilizes a simulated gacha-based manage system to introduce resources synthesis and randomized rewards. This simulates the unpredictability of wartime logistics while fostering player investment in the specific historical figures of Katipunan.

Content-Rich Campaign Progression; Provides structured missions strictly aligned with the factual timeline and key events of the Cavite Campaign, ensuring learners experience the history as a sequence of interconnected scenarios.

Kapatiran (Brotherhood) System: Emphasizes the real interpersonal dynamics among revolutionary forces by unlocking unique tactical synergies when specific characters support each other in combat.

Objectives

The general objective of this project is to develop Binakayan Rising, an isometric desktop role-playing game using the Unity engine, to provide an engaging computing solution that merges deployment-based strategic gameplay with historical education. This general objective serves as the overarching solution to the identified problem of the static historical instruction, ensuring the software acts as both a functional entertainment product and a rigorous educational tool.

Specifically, the project aims to:

1. Develop a Unity-based role-playing game that incorporates deployment-based, auto-battler mechanics, enabling players to strategically place historical units and observe AI-driven combat resolution
2. Implement a gacha-style recruitment and unit management system driven by resource synthesis and randomized rewards.
3. Design a structured campaign progression composed of missions reflecting key events of the Cavite Campaign.
4. Integrate an environmental and terrain system where battlefield conditions, such as trenches and coastal areas, mathematically influence combat outcomes.
5. Develop an interactive quiz feature that prompts player to answer historical questions during or after missions to reinforce knowledge.

Scope and Limitations

The general purpose of this study is to develop a functioning desktop prototype of "Binakayan Rising," an educational tactical role-playing game powered by the Unity engine. The target users and beneficiaries of this study are desktop gamers, students, and history enthusiasts. The period of the study will cover a standard capstone development cycle. The specific scope and features of the proposed software are the following:

1. **AI-Driven Auto-Battler Engine:** The core gameplay loop is divided into Deployment and Combat. During Deployment, players strategically place their recruited Katipunnan units onto valid 2.5D isometric grid tiles. Upon initiating the Combat phase, units autonomously navigate the terrain, target enemies, and calculate damage based on pre-programmed artificial intelligence scripts. This system also includes the Kapatiran (Brotherhood) mechanic, which calculates and applies unique tactical synergies when specific historical characters are deployed together.
2. **Simulated Gacha Module:** A character acquisition and management system driven by a Random Number Generation (RNG) algorithm, allowing players to pull historically inspired units and synthesize resources using earned in-game currency.
3. **The campaign** will be divided into sequential missions that represent major historical battles, strategic decisions, and important occurrences that took place in Cavite.
4. **Environmental Terrain System:** A mathematical modifier system granting defensive bonuses to units occupying specific tiles like earthworks or trenches.
5. **Interactive Pop-up Quiz:** Multiple-choice trivia regarding the Philippine Revolution that triggers at specific intervals and rewards the player with virtual currency upon current completion.

Limitations of the study

The specific boundaries and technical exclusions of the project are:

- A. **Hardware Dependency:** Graphical performance and algorithmic calculations are entirely dependent on the user's local hardware.
- B. **Strictly Offline Economy:** The gacha system operates on a closed local database. The software will explicitly not feature real-money microtransactions, e-commerce integrations, or live server economy synchronization.
- C. **Narrative Confinement:** Due to development time constraints, the educational content and 2.5D assets are strictly confined to the events of November 1896 and will not serve as a comprehensive encyclopedia of the entire revolution.
- D. **Network Isolation:** Developed exclusively as a single-player offline application with local JSON/SQLite saving, the system will not support multiplayer connectivity, PvP modes, cloud saving, or online leaderboards.

Review of Related Literature/Studies/Systems

Related Literature

According to Agoncillo (1990) in his foundational text *History of the Filipino People*, the victory at the Battle of Binakayan-Dalahican was heavily predicated on the advanced military engineering led by Edilberto Evangelista, specifically the construction of extensive trench networks that neutralized superior Spanish firepower. This supports the proposed system because it directly translates this historical literature into a mathematical gameplay mechanic, where placing units on "trench" tiles on the isometric grid grants defensive buffs.

Furthermore, historical memoirs such as those by Alvarez (1992) and the analysis of popular movements by Ileto (1979) highlight the deep interpersonal bonds and collective spirit that fueled the Katipunan forces. This relates to the proposed system because the software introduces the stylized Kapatiran relationship module to represent this dynamic, unlocking tactical synergies when specific characters support each other.

According to the *Handbook of Game-Based Learning* by Plass, Mayer, and Homer (2020), interactive digital environments are highly effective at scaffolding complex information by allowing learners to experiment with variables without real-world consequences. This supports the proposed system because it utilizes these modern GBL principles to transform the static history of the Cavite Campaign into a dynamic, safe interactive experience.

Expanding on educational game design, a systematic review by Clark, Tanner-Smith, and Killingsworth (2021) emphasizes that for a digital game to be an effective learning tool, its educational content must be seamlessly intertwined with its core progression mechanics. This relates to the proposed system because the interactive quiz module requires the player to answer historical pop-ups to earn the virtual currency needed for the gacha recruitment system, directly linking education to progression.

Finally, according to Schell (2019) in *The Art of Game Design*, randomized reward schedules and resource synthesis mechanics are among the most effective psychological tools for sustaining long-term player motivation in digital spaces. This supports the

proposed system's historically-driven recruitment and simulated gacha module. By translating the unpredictable nature of 1896 wartime logistics into a randomized reward system devoid of real-world monetary costs, the software ensures that players remain highly engaged with the progression loop and, consequently, the historical content.

Related Studies

Recent empirical studies analyzing deployment-based strategy games demonstrate that such environments highly stimulate predictive problem-solving and spatial reasoning. Research by Zhang, Cao, and Wang (2021) found that digital environments where users must strategically plan before a system executes a scenario yield significant improvements in student performance. This supports the proposed system because the core combat engine is built upon asynchronous mechanics, requiring players to analyze terrain and place units before the AI auto-battler takes over.

In the local context, Moscoso et al. (2020) demonstrated the effectiveness of immersive role-playing games in teaching Philippine revolutionary history, concluding that gamified, localized content deepens both cultural awareness and critical thinking. This supports the proposed system because the development builds directly upon these findings by applying modern game engine standards exclusively to the narrative of the Cavite Campaign.

A meta-analysis on the cognitive effects of gamification by Tokac, Novak, and Thompson (2022) revealed that Game-Based Learning environments significantly reduce the fear of failure, allowing students to repeatedly test hypotheses through trial and error. This relates to the proposed system because the auto-battler format allows the user to test different unit deployments; if a strategy fails, they can restart without real-world academic penalties.

Furthermore, a systematic review by Ravyse et al. (2022) identified that an intuitive, uncrowded user interface is paramount to prevent cognitive overload in serious games. This supports the proposed system because utilizing an asynchronous, deployment-based auto-battler removes the stress of high-speed, real-time clicking, allowing the user to focus entirely on spatial planning.

Additionally, research on embedded assessments in digital game-based learning by Shute

and Ke (2012) indicates that integrating educational quizzes directly into the core gameplay loop—rather than placing them as separate, post-game tests—significantly enhances knowledge retention. This directly supports the proposed system's interactive pop-up quiz feature. By interrupting specific intervals of the auto-battler combat to test the player's historical knowledge of the Cavite Campaign and immediately dispensing virtual currency upon a correct answer, the system effectively intertwines educational retention with gameplay.

Related Systems

Assassin's Creed Valhalla: Discovery Tour (Ubisoft, 2021) serves as a prime international example of a large-scale educational module, allowing users to explore historically accurate environments and interact with informational nodes. This relates to the proposed system because *Binakayan Rising* shares the same localized educational goal of providing a dedicated, interactive exploration of historical encampments.

Arknights (Hypergryph, 2020) provides an international standard for asynchronous, deployment-based combat and gacha roster management. This supports the proposed system because it adapts these highly engaging auto-deployment mechanics, shifting the context entirely from foreign fantasy settings to the historical reality of the 1896 Philippine Revolution.

Fire Emblem: Three Houses (Intelligent Systems, 2019) serves as a premier international benchmark for merging grid-based combat with relationship mechanics. The game utilizes a system where characters fighting in close proximity on the tactical grid build interpersonal bonds, subsequently unlocking narrative dialogue and mathematical combat buffs. This relates directly to the proposed system, as *Binakayan Rising* adapts this mechanical framework into the localized "Kapatiran" (Brotherhood) system, using grid deployment to mathematically represent the camaraderie of the Katipunan forces.

Teamfight Tactics (Riot Games, 2019) acts as a standard for modern auto-battler artificial intelligence and grid-based trait synergies. In the game, player input is locked during combat, and AI units navigate a grid to attack enemies based on pre-set deployment

positions and class synergies. This supports the proposed system by validating the technical feasibility of the AI-driven Auto-Battler engine and the Kapatiran mechanics running simultaneously within a unified logic loop.

In the national context, *Anito: Defend a Land Enraged* (Anino Entertainment, 2003) demonstrated early on that local myths and history could be successfully integrated into the role-playing game genre. This relates to the proposed system because *Binakayan Rising* modernizes this national effort by focusing purely on historical accuracy and integrating it with modern game design architectures.

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METHODOLOGY

Application of the Iterative Game Development Life Cycle (GDLC)

The researchers used an iterative Game Development Life Cycle (GDLC) as the software development methodology. This approach was selected because the system is composed of several connected modules (AI combat, gacha, UI, trivia) that require repeated planning, risk analysis, testing, and refinement. Through this iterative model, the researchers developed the system by cycle, identifying potential logical errors in pathfinding and balancing issues early on.

- **Phase 1: Concept, Planning, and Requirement Analysis:** In the first phase, the researchers established the core mechanics (deployment-based auto-battler, gacha, quizzes) and identified the target users. A key risk involved inaccurate historical translation. This was mitigated by conducting formal data gathering and securing validation from a Subject Matter Expert (SME).
- **Phase 2: Architectural Design and Asset Pipeline:** In the second phase, the researchers formalized the system through standard software engineering models (Data Flow Diagrams, Entity-Relationship Diagrams, Finite State Machines). Concurrently, graphical assets for the 2.5D isometric terrain and character sprites were mapped out.
- **Phase 3: Core Module Development (Combat & Terrain):** The third phase focused heavily on programming the AI-driven auto-battler and the environmental terrain system. A major risk was pathfinding errors on the isometric grid. This was addressed by utilizing Unity's built-in physics and strict orthographic grid snapping to ensure reliable unit navigation.
- **Phase 4: Economy and UI Integration (Gacha & Quizzes):** The fourth phase involved integrating the localized RNG gacha module and the interactive pop-up quizzes. Risks included UI cognitive overload and broken save states. These were addressed by keeping the combat UI minimal and strictly managing data persistence through local SQLite databases.
- **Phase 5: Testing, Evaluation, and Refinement:** In the final phase, the system underwent rigorous playtesting to balance unit statistics, Kapatiran synergies, and hardware performance.

Technical Background

Technologies to be Used

The development of *Binakayan Rising* relies on current trends in independent video game development, prioritizing accessible but powerful object-oriented environments.

The primary technology utilized is the Unity Engine 6.4, selected specifically for its capability in seamlessly rendering 2D sprites within 3D scenarios, which is critical for the game's 2.5D isometric aesthetic, and its robust tools for driving the automated battle system. The core logic, including the asynchronous AI pathfinding algorithms, Kapatiran synergy modifiers, and the simulated gacha RNG module, is programmed using the C# language via Microsoft Visual Studio.

To manage persistent offline data—such as tracking virtual currency, unit rosters, and campaign progression—without requiring cloud server integration, the system utilizes local JSON serialization and a lightweight SQLite database. Graphical assets, including UI elements, isometric grids, and character portraits, are generated using raster graphics editors like Aseprite and Adobe Photoshop.

Calendar of Activities

The development of the project is structured chronologically from January to April 2026. The initialization phase began in late January with group formation and project orientation. In February, the team dedicated the first week entirely to project conceptualization and brainstorming, transitioning into the drafting of the manuscript and early system prototyping. March was dedicated to rigorous consultation and core development. The team conducted semi-structured interviews with the historical Subject Matter Expert to validate the scope, which ran concurrently with the programming of the isometric grid and user interface. Finally, April was allocated for system validation, playtesting the combat mechanics, addressing initial project feedback, and finalizing manuscript revisions for the final defense. For a comprehensive visual breakdown of these milestones, task allocations, and specific durations, refer to Appendix C: Calendar of Activities.

Resources

To successfully develop, test, and implement the *Binakayan Rising* desktop application, the proponents require specific hardware and software resources capable of handling game engine compilation and digital asset creation.

For hardware, the primary requirements include development desktop computers or laptops running Windows 10 or higher. These machines must be equipped with multi-core processors (e.g., Intel Core i5 or AMD Ryzen 5 equivalents), a minimum of 8GB to 16GB of RAM, and dedicated Graphics Processing Units (GPUs) to ensure stability while rendering visual effects and running physics calculations within the Unity editor. Additionally, graphic drawing tablets will be utilized by the proponents to digitally illustrate the 2D character portraits and UI elements. A secondary, lower-specification desktop will also be required as a dedicated testing environment to accurately evaluate the game's minimum hardware requirements.

In terms of software resources, the core application required is the Unity Engine for 2D rendering and state-machine management. Microsoft Visual Studio will serve as the Integrated Development Environment (IDE) for writing and debugging the C# scripts. Raster graphics editors like Adobe Photoshop and Aseprite will be employed for art creation. Furthermore, to facilitate collaborative code development, the proponents will use Git paired with GitHub for remote revision control, alongside agile project management software such as Trello to track milestone progress.

Requirements Analysis

To accurately determine the pedagogical and mechanical requirements of the system, a qualitative data-gathering procedure was conducted. The primary data gathering tool utilized by the researchers was a Key Informant Interview Questionnaire. A semi-structured consultation was conducted directly with a recognized Subject Matter Expert (SME) in Philippine History to understand the historical context and how to translate it into gameplay.

This direct consultation allowed the researchers to gather specific expectations for the proposed game, validating the appropriateness of integrating General Edilberto Evangelista's trench warfare tactics into an auto-battler format. The SME confirmed that the asynchronous deployment mechanics successfully stimulate analytical thinking, while the simulated gacha and trivia components effectively engage the target youth demographic. To ensure the authenticity and validity of the gathered data, the interview transcripts were formally reviewed and signed by the historical consultant, and an official Certificate of Validation was secured.

Requirements Documentation

The proposed *Binakayan Rising* system is expected to provide a highly structured, offline digital experience. The system requires secure local data persistence to track the "Tactician's" (the user's) progress, including accumulated Reales (currency), Rations (energy), and individual unit experience levels.

The system provides a distinct Encampment UI acting as the hub world, requiring functionality for resource generation (Farms/Mines), weapon upgrades (Synthesis), and character roster management. Upon entering a combat scenario, the system strictly enforces a phased approach. During the *Deployment Phase*, the system requires drag-and-drop mechanics mapped to a 2.5D isometric grid, allowing the player to position units. During the *Combat Phase*, the system requires user input to be locked while the AI engine calculates autonomous pathfinding, damage algorithms, and Kapatiran synergy triggers based on proximity. Finally, the system requires an interruptive trivia module that evaluates user input against a database of historical facts to dispense rewards.

Design of Software, System, Product, and/or Processes

The system architecture of *Binakayan Rising* is mapped using standard UML methodologies to ensure a robust and modular development process.

Use Case Diagram The Use Case Diagram models the dynamic interactions between the system's actors and its core functionalities. To ensure a clear boundary between human input and automated logic, the diagram defines two primary actors: the **Player (Tactician)** and the **Game AI System**.

- **Primary Actor (The Player):** The Player is the user interacting with the graphical interface. Their use cases are strictly limited to pre-combat and mid-combat interactions, including: *Manage Encampment* (collecting resources), *Summon Heroes* (triggering the RNG gacha), *Synthesize Equipment*, *Deploy Units* (dragging units to the grid), and *Answer Historical Quiz*.
- **Secondary Actor (Game AI System):** The Game AI System handles all autonomous calculations once the player locks in their deployment. Its use cases operate independently of player input, including: *Execute Combat*, *Calculate Pathfinding*, *Apply Kapatiran Synergy* (checking proximity for buffs), *Calculate Damage*, and *Dispense Rewards*.

State Machine Diagram To govern the overarching application flow and prevent memory overload during intense AI combat calculations, the system utilizes a Finite State Machine (FSM) architecture managed by the Unity Game Manager. The system transitions through five distinct states:

1. **Main Menu State:** The entry point of the application. The system listens for the user to either "Start New Campaign" or "Load Local Save." Upon selection, it transitions to the Encampment State.
2. **Encampment Management State:** The primary idle state where the user interacts with the hub UI (Farms, Mines, Hero Summoning, and Synthesis). Memory allocation is focused purely on UI interactions and database querying. Selecting the 'Mission Portal' triggers a transition to the Pre-Battle Deployment State.
3. **Pre-Battle Deployment State:** The system loads the 2.5D isometric map. The state listens for mouse drag-and-drop inputs to place units. Once the player clicks "Start Battle," the system locks all user inputs and transitions to the Auto-Combat State.
4. **Auto-Combat State:** The core algorithmic state. The UI is hidden, and the system loops through AI pathfinding and damage calculations. If all enemies are routed or player units defeated, it transitions back to the Encampment State. If a mid-battle interval is reached, it transitions to the Quiz State.

5. **Paused Quiz State:** Combat physics and timers are frozen while the trivia UI is displayed. Once an answer is submitted and rewards calculated, it transitions back to the Auto-Combat State.

Entity-Relationship Diagram (ERD) Finally, the offline data architecture is structured using an Entity-Relationship Diagram. The ERD maps the primary relationships between the User Profile (tracking accumulated Reales, Rations, and campaign progress), the Unit Roster (storing individual hero statistics, Kapatiran synergies, equipped weapons, and experience levels), and the Inventory Ledger.

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APPENDICES

APPENDIX A. TABLES

Component	Minimum Requirement (End-User)	Recommended Requirement (Development)
Operating System	Windows 10 (64-bit)	Windows 10 / 11 (64-bit)
Processor	Intel Core i3 or AMD Ryzen 3 equivalent	Intel Core i5 / AMD Ryzen 5 or higher
Memory (RAM)	8 GB RAM	16 GB RAM
Graphics (GPU)	Integrated Graphics (Intel UHD) or 2GB VRAM	Dedicated GPU (NVIDIA/AMD) with 4GB+ VRAM
Storage Space	2 GB available space	10 GB available space (SSD recommended)
Input Devices	Keyboard and Mouse	Keyboard, Mouse, and Graphic Drawing Tablet
Software Stack	SQLite (Local Database)	Unity Engine 6.4, Microsoft Visual Studio (C#)
Asset Creation	N/A	Aseprite, Adobe Photoshop, GitHub (Version Control)

Table 1. Hardware and Software Requirements

Terrain Type	Visual Description	Mathematical Effect / Modifier
Standard Grid	Flat grass, dirt paths, or clearing	None (Uses base unit statistics)
Evangelista's Trench	Dug-out earthworks with bamboo	+20% Defense, +15% Evasion
Coastal Shallows	Water tiles near the Dalahican shore	-15% Movement Speed, - 10% Defense
Bamboo Barricade	Sharp wooden defensive structures	Impassable (Blocks enemy pathfinding)
Encampment Tent	Safe zones at the rear of the grid	+5% HP Regeneration per AI turn

Table 2. Environmental Terrain Modifiers

Character Pair	Support Rank C	Support Rank B	Support Rank A (Maximum)
Gen. Evangelista & Emilio Aguinaldo	Unlocks Lore Dialogue 1	+10% Attack Damage	+15% Attack Damage, +10% Defense
Katipunero Vanguard & Field Medic	Unlocks Lore Dialogue 2	+15% Healing Received	+25% Healing Received, +5% Max HP
Caviteño Marksman & Trench Engineer	Unlocks Lore Dialogue 3	+10% Ranged Accuracy	+20% Ranged Accuracy, +1 Attack Range
Magdalo Infantry & Magdiwang Infantry	Unlocks Lore Dialogue 4	+5% Critical Hit Chance	+15% Critical Hit Chance, +10% Evasion

Table 3. Kapatiran (Brotherhood) Synergy Levels

Historical Question	Multiple Choice Options	Correct Answer	In-Game Reward
Who was the primary engineer responsible for designing the trench networks in Cavite?	A. Andres Bonifacio B. Emilio Aguinaldo C. Edilberto Evangelista D. Antonio Luna	C. Edilberto Evangelista	+50 Reales, Map- wide +10% HP Heal
When did the simultaneous Battle of Binakayan- Dalahican take place?	A. August 1896 B. November 9-11, 1896	B. November 9-11, 1896	+50 Reales, +10% Attack Buff for 1 turn

	C. June 12, 1898 D. December 30, 1896		
Who was the Spanish Governor-General that launched the attack on Cavite during this battle?	A. Ramón Blanco B. Camilo de Polavieja C. Miguel López de Legazpi D. Fernando Primo de Rivera	A. Ramón Blanco	+50 Reales, Resets AI Enemy positions
The victory at Binakayan-Dalahican is historically recognized as the first major Filipino victory in which province?	A. Manila B. Bulacan C. Laguna D. Cavite	D. Cavite	+50 Reales, Instantly revives 1 fallen unit

Table 4. Sample Historical Trivia Database (Educational Module)

Evaluation Criteria	Description for <i>Binakayan Rising</i>	Target Measurement
Functional Suitability	The degree to which the deployment grid, AI auto-battler, and gacha mechanics function exactly as designed without logical errors.	100% of core mechanics execute without game-breaking bugs.
Performance Efficiency	The ability of the Unity Engine to render 2.5D graphics and calculate AI pathfinding without dropping frame rates on minimum hardware.	Maintains a stable 30 to 60 Frames Per Second (FPS) during combat.
Usability (UI/UX)	The ease with which users can navigate the encampment menus, read the historical trivia, and drag-and-drop units onto the grid.	Positive feedback on UI clarity from 85%+ of survey respondents.
Reliability	The software's ability to save and load offline progress (JSON/SQLite) accurately without corrupting the player's roster or currency.	Zero instances of corrupted save files during the testing phase.

Table 5. Software Evaluation Criteria (ISO 25010 Standard)

APPENDIX B. FIGURES

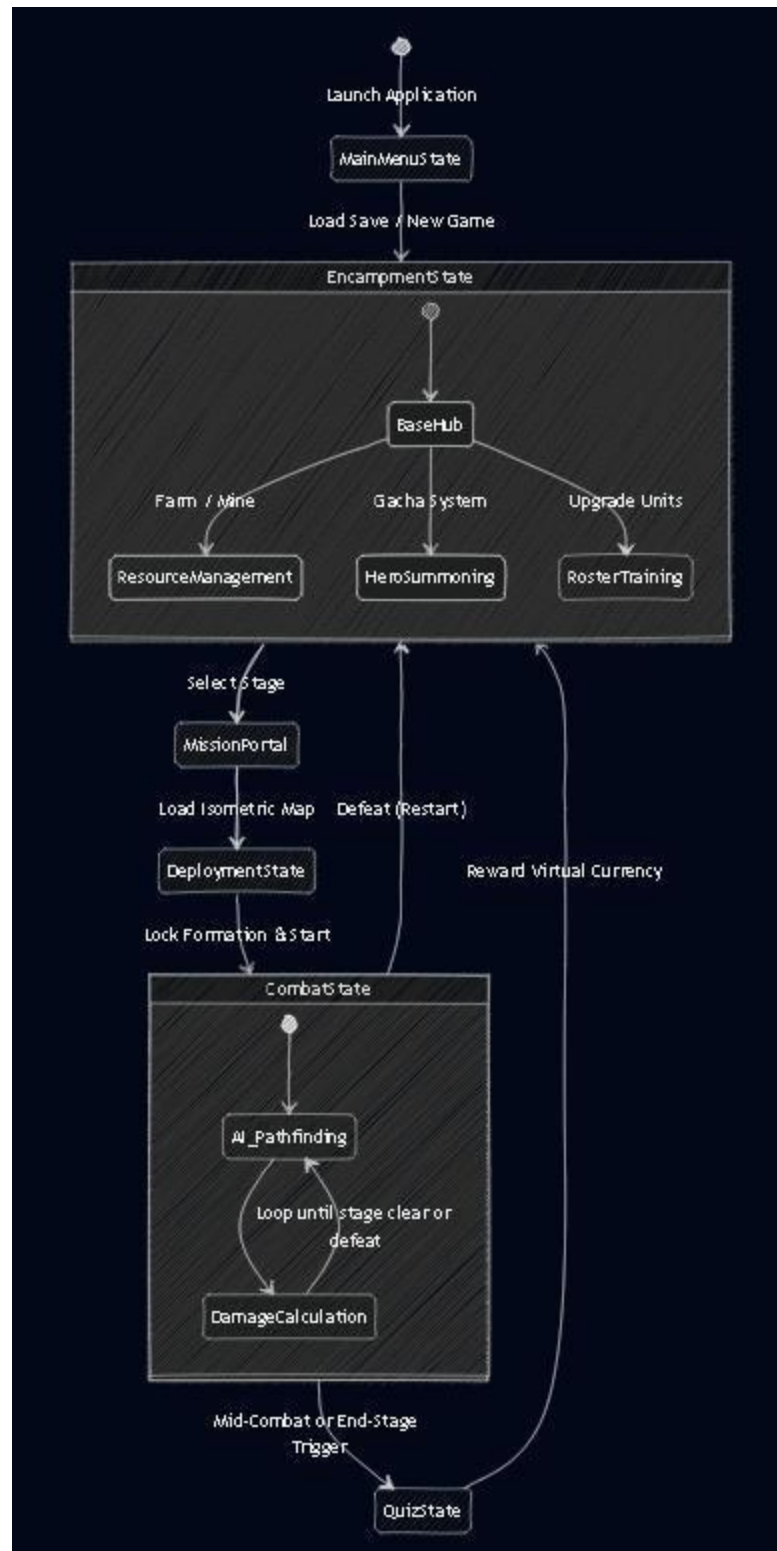


Figure 2. State Machine Diagram

APPENDIX C. GANTT CHART

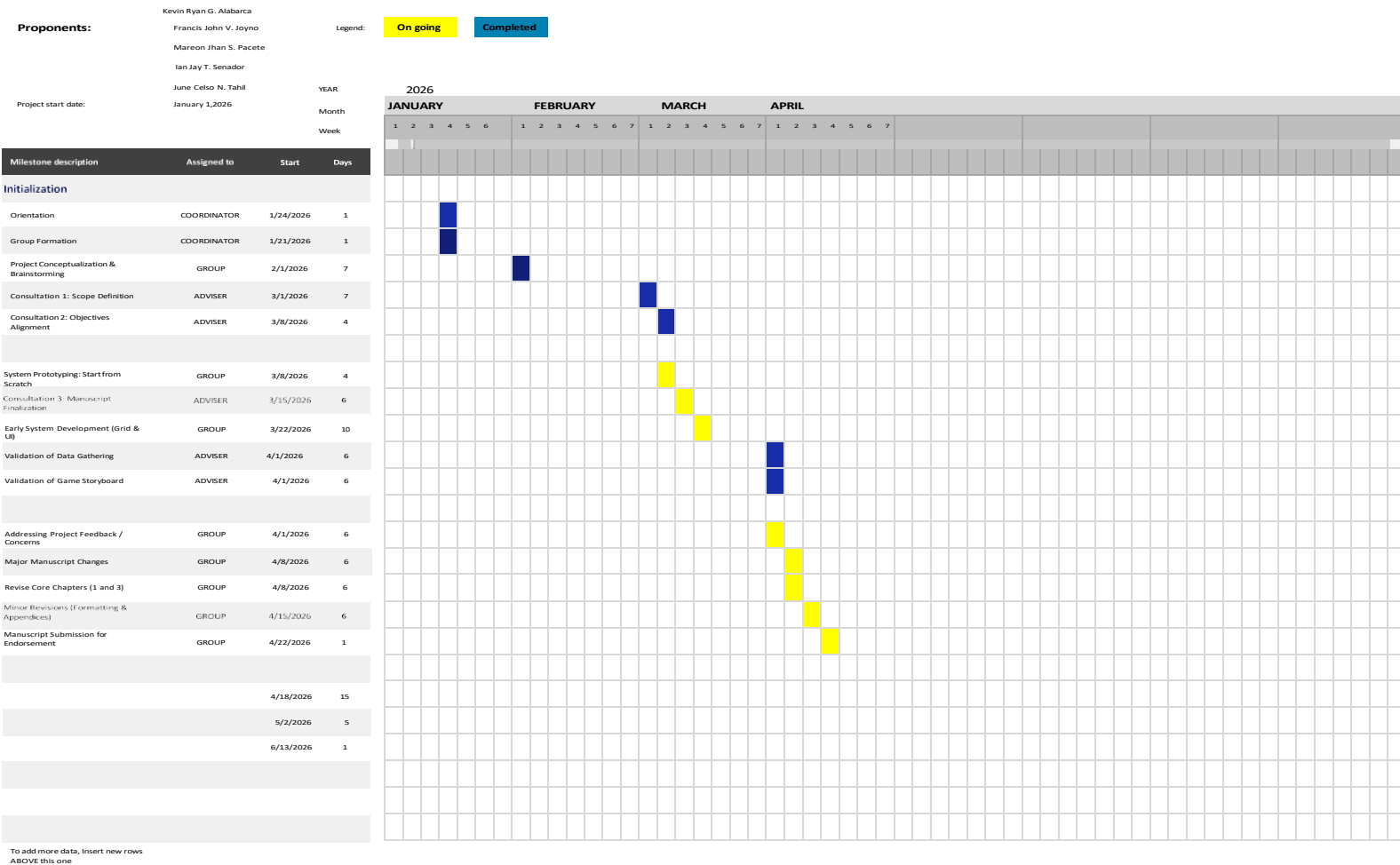


Table 6: Calendar of Activities (Gantt Chart)

APPENDIX D. EVALUATION FORMS/TOOLS

COVER LETTER TEMPLATE

March 27, 2026

Adnan RazulHistorian/ MSU Gensan Faculty
STI College- Gen. Santos, Inc.
General Santos City

Dear Mr. Razul,

Mabuhay!

The undersigned are 3rd-year Bachelor of Science in Information Technology (BSIT) students currently working on our capstone project titled **"Binakayan Rising: An Isometric Desktop Role-Playing Game Based on the Historical Battle of Binakayan–Dalahican"**. This study aims to develop a desktop application that merges tactical, deployment-based gameplay with educational historical content to promote local history among the youth.

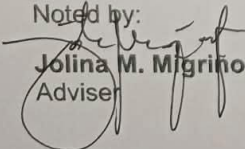
In connection with this, we would like to respectfully ask for your permission, time, and expertise to serve as our Subject Matter Expert (SME). We request your participation in a short interview to validate the historical accuracy of our Game Design Document, specifically regarding General Edilberto Evangelista's engineering tactics and the events of the 1896 Cavite Campaign.

Rest assured that your insights and any data gathered from this consultation will be used strictly for academic and development purposes and will be treated with the utmost respect

Respectfully yours,

Mareon Jhan S. Pacete/09700718420

Noted by:


Jolina M. Migrino, LPT
Adviser

\

APPENDIX E. RESOURCE PERSON



March 27, 2026

Andan Razul
Historian
Mindanao State University
General Santos City

Dear Sir/Ma'am,

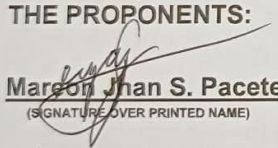
Mabuhay!

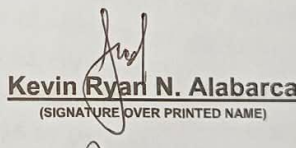
The undersigned is currently working on research titled "*Binakayan Rising: An Isometric Desktop Role-Playing Game Based on the Historical Battle of Binakayan-Dalahican.*" This study aims to gather data related to the effective method for teaching abstract and heavily narrative-driven subjects like history, transforming traditional education away from passive absorption toward active student engagement. The result of this study will be used to develop "*Binakayan Rising*" an offline, isometric 2.5D desktop role-playing game.

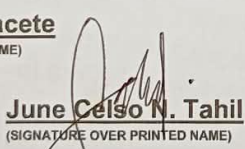
In connection with this, we would like to ask your permission and endorsement to allow us to conduct survey/interview the Andan Razul. Rest assured that whatever data will be gathered from this study will be treated with utmost confidentiality.

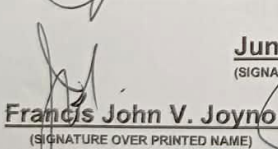
I am hoping for your favorable response to this undertaking.

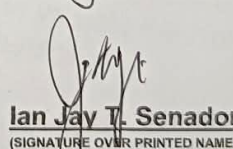
Respectfully yours,
THE PROPONENTS:


Marlon Jhan S. Pacete
(SIGNATURE OVER PRINTED NAME)

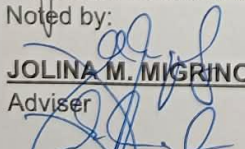

Kevin Ryan N. Alabarca
(SIGNATURE OVER PRINTED NAME)


June Celso N. Tabil
(SIGNATURE OVER PRINTED NAME)

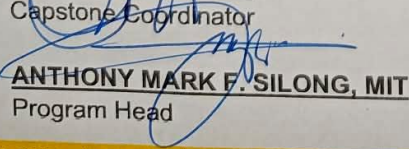

Francis John V. Joyno
(SIGNATURE OVER PRINTED NAME)


Ian Jay T. Senador
(SIGNATURE OVER PRINTED NAME)

Noted by:


JOLINA M. MIGRINO
Adviser


JOLINA M. MIGRINO
Capstone Coordinator


ANTHONY MARK F. SILONG, MIT
Program Head

APPENDIX F. GAME DESIGN DOCUMENT

Game Name: Binakayan Rising Genre: Tactical Role-Playing Game (TRPG) / Deployment-Based Auto-Battler Player: 1 (Single-player, Offline) Game Elements: Grid-based tactical deployment, base management, resource gathering, offline gacha hero summoning, interactive historical pop-up quizzes, and the Kapatiran (Brotherhood) Synergy System.

TECHNICAL SPECIFICATION

- Technical Form: 2.5D Graphics (2D character sprites rendered within a 3D isometric environment).
- View: Isometric Top-Down view for combat; Wide-angle flat view for base management.
- Platform: Windows PC.
- Software: Unity Engine 6.4, Microsoft Visual Studio (C#), Aseprite / Adobe Photoshop, JSON/SQLite (Local Database).
- Language: C#
- Device: Desktop / Laptop PC (Keyboard and Mouse).

GAME PLAY

Binakayan Rising merges a rich, biographical narrative with deep, mathematical tactical systems. The game is an asynchronous, deployment-based auto-battler set during the 1896 Philippine Revolution, specifically chronicling the rise of General Edilberto Evangelista.

The Lore: The Architect's Ascent

- Act 1: The Scholar of Ghent (The Prologue) – The narrative opens in the quiet halls of Ghent University in Belgium. Edilberto Evangelista is a top-tier civil engineering student living a life of comfort. However, smuggled letters detailing the outbreak of the Katipunan revolution reach him. Forced to choose between a prosperous future in Europe and a perilous return to his homeland, Evangelista packs his blueprints and boards a ship to Manila.
- Act 2: The Catalyst of Defeat (The Awakening) – Arriving in Cavite, Evangelista is initially dismissed by veteran commanders who believe wars are won through raw passion and close-quarters combat. Relegated to an observer, he witnesses a tragic skirmish where passionate Filipino fighters charge across an open field, only

to be slaughtered by modern Spanish artillery. This tragedy cements a harsh reality: raw courage is suicidal without modern military engineering.

- Act 3: Forging the Brotherhood (The Rise to Command) – Bypassing the old guard, Evangelista builds a hidden encampment, recruiting farmers, teachers, and medics to forge a disciplined "Kapatiran" (Brotherhood). Recognizing his logistical brilliance, the Katipunan leadership grants him command. Evangelista immediately begins transforming the geography of Cavite, designing sophisticated trench networks and bamboo fortifications.
- Act 4: The Masterpiece of Binakayan-Dalahican (The Climax) – The narrative peaks between November 9 and 11, 1896. Governor-General Ramón Blanco launches a massive, synchronized land and naval armada. The player is tasked with executing Evangelista's grand design, deploying units into meticulously crafted earthworks to secure the historical victory at Binakayan.

Game Play Outline:

- Opening the game application: The player is greeted by a high-definition 2.5D splash screen of Katipunan forces in the Cavite trenches, with options to start a new campaign or load a local save.
- Game options: Audio toggles for background music and sound effects, resolution settings, and a historical glossary tab.
- Modes: Linear Historical Campaign Mode.
- Game Elements: Terrain defense modifiers, randomized gacha hero acquisition, interactive trivia, and the "Kapatiran" relationship-building system.
- Game Quests / Levels: The campaign is divided into structured quests aligned with the historical timeline:
 - *Level 1: The Architect's Awakening* (Sub-quests: The Scholar of Ghent [UI Tutorial], The Reality Check [Combat Tutorial], The Vanguard's Rally [Base Management], Scavenging for the Cause [Resource Gathering]).
 - *Level 2: Preparations for War* (Sub-quests: The Intercepted Armada [Synergy Drill], Forging the Earthworks [Defense/Escort], The Silent Sabotage [Stealth/Intelligence]).

- *Level 3: Battle of Binakayan-Dalahican* (Sub-quests: The First Wave [Nov 9], The War of Attrition [Nov 10], The Decisive Dawn [Nov 11]).
- Player controls: Mouse-driven interface (Point-and-click to navigate the base; Drag-and-drop to place units during deployment).
- Winning: The AI successfully routs all enemy units on the isometric grid based on the player's strategic deployment, alongside correctly answering educational quizzes.
- Losing: The player's deployed roster is completely defeated by the Spanish AI, forcing a return to the encampment to rethink their formation, upgrade weapons, or recruit new units.
- End: A visual-novel style victory sequence detailing the historical aftermath of November 11, 1896, followed by a return to the Encampment screen.
- Why is all this fun?: It merges the highly addictive progression loop of gacha recruitment and base building with the deep, rewarding intellectual puzzle of spatial deployment, all while gamifying local history.

DESIGN DOCUMENT

Design Guidelines The primary design goal is to translate the static historical facts of the Battle of Binakayan-Dalahican into dynamic, engaging gameplay mechanics without relying on real-money microtransactions. Creative restrictions dictate that all weapons, terrain, and trivia must adhere strictly to the 1896 historical context. The system must operate entirely offline, utilizing local data storage.

Game Design Definitions The focus of the gameplay is tactical positioning and resource economy. Because combat resolves autonomously, the player's primary intellectual challenge occurs *before* weapons are drawn. A player wins a combat instance by successfully predicting enemy AI pathfinding and placing their Vanguard and Marksman units in optimal terrain.

Game Flowchart (Mechanics Integration)

- Menu: Handles local saved JSON/SQLite data instantiation and audio preferences.
- Synopsis: Visual novel dialogue system triggers upon entering Hero Camps to advance the Evangelista lore.

- **Game Play:** The Unity Game Manager toggles the Finite State Machine between the Base Management State, Pre-Battle Deployment State, and Auto-Combat State.
- **Player Control:** Raycasting from the mouse pointer detects grid tiles for unit placement and UI hitboxes.
- **Game Over (Winning and Losing):** The system calculates remaining unit HP at the end of every AI turn to trigger the Victory/Defeat screen overlays.

Player Definition The player acts as the unseen "Supreme Tactician" of the Katipunan forces. They do not possess a physical avatar on the battlefield but act as the overarching hand that guides resource allocation in the encampment and dictates unit starting positions during skirmishes.

Player Properties:

- **Reales (Virtual Currency):** Used for the Hero Summoning gacha. *Feedback:* UI counter increments with a coin-clinking sound effect when quizzes are answered correctly.
- **Rations (Energy):** Required to deploy units into the Mission Portal. *Feedback:* Depletes upon entering a stage; UI text flashes red if insufficient.
- **Scrap Metal (Crafting):** Used in the Synthesis facility. *Feedback:* Gathered over time from the Mines, updating the inventory ledger.

Player Rewards (Power-ups and Pick-ups)

- **Pop-up Quiz Bonuses:** Correctly answering mid-battle historical trivia instantly rewards the player with bonus Reales and triggers a "Tactician's Command" effect—applying a temporary map-wide HP heal or ATK buff to surviving Katipunan units.
- **Kapatiran (Brotherhood) Support Ranks:** Placing specific historical units adjacent to each other on the deployment grid builds their Support Levels (Ranks C, B, and A). Maximizing this meter rewards the player with unique dialogue scenes and permanent evasion/accuracy stat boosts when those units fight together.
- **Synthesized Weapons:** Combining scrap metal rewards the player with upgraded equipment (e.g., transitioning a unit from wielding a standard bolo to a captured Mauser rifle), significantly increasing base damage output.

User Interface (UI) The game relies entirely on a streamlined PC interface optimized for the mouse, designed to prevent cognitive overload.

- Left-Click: Select a unit, interact with a facility in the encampment, confirm a drag-and-drop grid placement during deployment, or select a quiz answer.
- Right-Click: Cancel the current action, deselect a unit, or close a pop-up menu.
- Scroll Wheel: Zoom the camera in and out of the isometric battlefield or the wide-angle encampment.
- Deployment UI: During the pre-battle phase, the UI displays the player's available roster at the bottom of the screen. The player clicks and drags hero portraits onto highlighted blue valid grid tiles.
- Combat UI: Once "Start Battle" is clicked, all deployment UI fades away to provide an unobstructed view of the AI-driven skirmish, showing only floating health bars above the 2D sprites.
- Esc Key: Pauses the game, opening the settings and exit prompts.

APPENDIX G. PROOF OF DATA GATHERING

APPENDIX H. TRANSCRIPTS OF PRELIMINARY HISTORICAL CONSULTATION

Description: To ensure the educational direction of *Binakayan Rising* was fundamentally sound before full development, a preliminary consultation was conducted with a subject matter expert in history. The following table contains the verbatim audio transcriptions gathered during this conceptual evaluation. The raw transcripts preserve the original language (Taglish) spoken by the expert to ensure data authenticity. An English translation is provided for formal manuscript integration.

Interviewer: What are your thoughts on the proposed historical concept and its overall educational value for the players?

Subject Matter Expert: "In my opinion, this proposed game—especially given its historical focus on the Battle of Binakayan-Dalahican—provides incredible value. Rather than treating Philippine revolutionary history as a static textbook lesson, it transforms it into an active, immersive experience. It allows players to better understand our cultural history while simultaneously enjoying the narrative, seamlessly masking the educational aspects behind engaging gameplay."

Interviewer: How do you feel about the intended gameplay mechanics, specifically the grid system and combat execution?

Subject Matter Expert: "The game's concept is excellent because it demands genuine analytical thinking and helps us recall our history. The planned mechanics will genuinely challenge you to think strategically about placing characters in their optimal positions on the grid. In a tactical setup like this, especially with an automated battle system, the player's pre-combat unit placement and understanding of terrain will directly dictate the outcome, which provides the exact intellectual challenge players look for."

Interviewer: Do you think the additional features, like the quizzes and gacha mechanics, will effectively appeal to your target audience?

Subject Matter Expert: "Absolutely. Through this proposed game, players can explore and enhance their knowledge, particularly in history. The inclusion of puzzle elements

and planned quizzes act as interactive checkpoints to reinforce historical facts without breaking the game's momentum. Furthermore, integrating a gacha system—where players might pull or unlock different historical figures, revolutionary units, or authentic period weaponry—makes the concept highly engaging and creates a compelling long-term progression loop, especially for today's youth."

Interviewer: Finally, what is your assessment of the visual and technical design, including the user interface?

Subject Matter Expert: "Overall, the proposed design is well-crafted. I really appreciate the conceptual graphics—blending expressive 2D character sprites within detailed 3D isometric scenarios creates a visually striking desktop experience. I also want to highlight the UI and the planned balancing for character placement. It will certainly require a clean interface to effortlessly communicate complex stats, grid advantages, and unit synergies so players can focus entirely on the analytical thinking needed to defeat their opponents."

APPENDIX I. REVISION LIST

Binakayan

REVISIONS LIST

REVISIONS LIST

The following revisions need to be complied with by the Capstone Project Group. Further evaluation of the capstone project paper will depend on the indicated result below.

CAPSTONE PROJECT PART/ CHAPTER	PAGE NUMBER	LIST OF REVISIONS
Requirements Analysis	13	→ list and discuss the data gathering procedures / tools used in the study → some parts of the docs are not well-designed → revise the use-case diagrams → transcript of interview should be signed by the consultant → ask for certificate from your consultant → specify the guests and subguests in GDD ↳ auto button → specify the lore / theme

REVISIONS LIST

[illegible]

- follow the APA 7th edition format.

Binakayan

REVISIONS LIST

The following revisions need to be complied with by the Capstone Project Group. Further evaluation of the capstone project paper will depend on the indicated result below.

[illegible]

APPENDIX J. FORMS/DOCUMENTS

ADVISER'S ACCEPTANCE FORM

Group Members: ALABARCA, Kevin Ryan
JOYNO, Francis John
PACETE, Mareon Juan
SENADOR, Ian Jay
TAHIL, June Celso

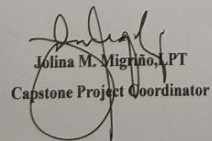
APPROVED RESEARCH TITLE: Binakayan Rising: An Isometric Desktop Role-Playing Game
Based on the Historical Battle of Binakayan-Dalahican

AREA OF STUDY:

CONFORMED:

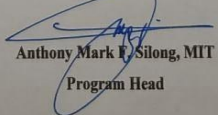

Jolina M. Migrino, LPT
Adviser's Name

APPROVED BY:


Jolina M. Migrino, LPT
Capstone Project Coordinator

Date: 03/23/2026

NOTED BY:

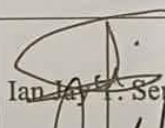
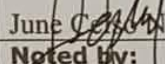
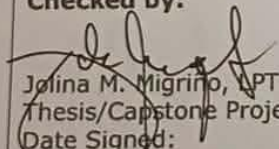
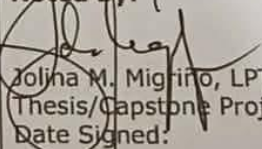

Anthony Mark F. Silong, MIT
Program Head

APPENDIX K. ACCOMPLISHMENT AND COSLTATION FORMS

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title: BINAKAYAN RISING: AN ISOMETRIC DESKTOP ROLE-PLAYING GAME BASED ON THE HISTORICAL BATTLE OF BINAKAYAN-DALAHICAN
Week Number: 1

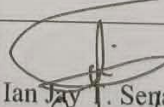
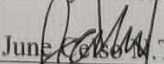
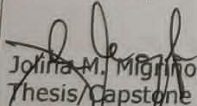
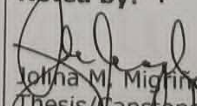
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
• Orientation & Group Formation: Attended the official Capstone Project orientation. Formed the project group and assigned technical roles (e.g., Lead Programmer, UI/UX Designer, Game Artist, Documentation Lead). • Brainstorming: Conceptualized the idea for a local historical game focusing on the Philippine Revolution.	
Prepared by: Kevin Ryan N. Alabarca Francis John V. Joyno Mareon S. Pacete	 Ian Jay T. Senador  June Carlo M. Tabil
Checked by:  Jolina M. Migrino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migrino, LPT Thesis/Capstone Project Coordinator Date Signed:

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Thesis/Capstone Project Title: BINAKAYAN RISING: AN ISOMETRIC DESKTOP ROLE-PLAYING GAME BASED ON THE HISTORICAL BATTLE OF BINAKAYAN-DALAHICAN
Week Number: 2

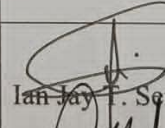
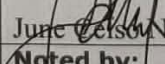
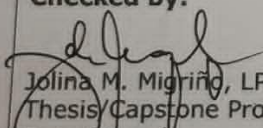
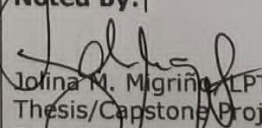
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
• Project Title Proposal: Finalized the scope to focus on General Edilberto Evangelista and the Battle of Binakayan-Dalahican. Drafted the official Title Proposal. • Adviser Selection: Officially selected and secured the Adviser's Acceptance Form. • Game Design Document (GDD) Draft: Started drafting the initial GDD, outlining the Deployment-Based Auto-Battler mechanics and the offline Gacha system.	
Prepared by: Kevin Ryan N. Alabarca Francis John V. Joyno Mareo S. Pacete	 Ian Jay T. Senador  June Teodoro M. Tabil
Checked by:  Jolina M. Migfino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migfino, LPT Thesis/Capstone Project Coordinator Date Signed:

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Thesis/Capstone Project Title: BINAKAYAN RISING: AN ISOMETRIC DESKTOP ROLE-PLAYING GAME BASED ON THE HISTORICAL BATTLE OF BINAKAYAN-DALAHICAN
Week Number: 3

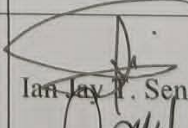
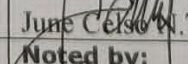
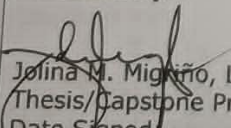
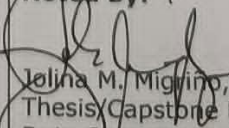
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Manuscript Drafting (Chapters 1 & 2): - Drafted the Project Context, identifying the gap in localized educational games regarding Philippine history. - Drafted the Review of Related Literature (RRL) and identified international and national games for reference. - Established the Game Development Life Cycle (GDLC) as the methodology.	
Prepared by: Kevin Ryan N. Alabarca Francis Joy V. Joyno Mareon S. Pacete	 Ian Jay T. Senador  June Celsa N. Tahil
Checked by:  Jolina M. Migrino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migrino, LPT Thesis/Capstone Project Coordinator Date Signed:

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Thesis/Capstone Project Title: BINAKAYAN RISING: AN ISOMETRIC DESKTOP ROLE-PLAYING GAME BASED ON THE HISTORICAL BATTLE OF BINAKAYAN-DALAHICAN
Week Number: 4

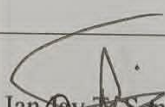
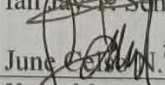
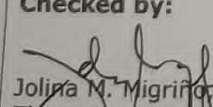
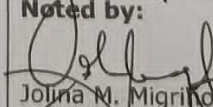
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Development Environment Setup: - Configured the Unity Engine 6.4 workspace and set up GitHub for collaborative version control. - Designed the initial architectural diagrams (Use Case, Entity-Relationship, and State Machine diagrams) for Chapter 3. - Prepared the Key Informant Interview questionnaire and formal letter to be sent to the historical Subject Matter Expert (SME).	
Prepared by: Kevin Ryan N. Alabarca Francis Joy V. Joyno Marlon S. Pacete	 Ian Jay T. Senador  June Celso M. Tabil
Checked by:  Jolina M. Miglino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Miglino, LPT Thesis/Capstone Project Coordinator Date Signed:

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Week Number: 5

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
On-Site Data Gathering & Consultation: - Conducted the formal interview with the Historical Subject Matter Expert (SME). - Validated the translation of the historical trench warfare into mathematical game modifiers. - Gathered primary data to construct the historical pop-up quizzes. - Secured the SME's signature on the Interview Transcript for documentation.	
Prepared by: Kevin Ryan N. Alabarca Francis John V. Joyno Mareon S. Pacete	 Ian Jay B. Sonador  June C. Tahil
Checked by:  Jolina M. Migrino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migrino, LPT Thesis/Capstone Project Coordinator Date Signed:

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Week Number: 6

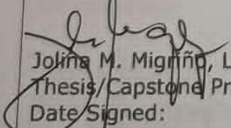
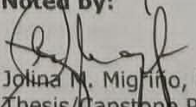
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Core Module Development (Deployment & Terrain): - Programmed the base 2.5D isometric grid system in Unity. - Implemented mouse drag-and-drop mechanics for pre-battle unit deployment. - Coded the terrain system, applying +20% Defense and Evasion modifiers to units placed on "Evangelista's Trench" tiles.	
Prepared by: Kevin Ryan N. Alabarca Francis John V. Joyno Mareon S. Pacete	Ian P. Senador June Celso N. Tabil
Checked by: Jolina M. Migriño, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by: Jolina M. Migriño, LPT Thesis/Capstone Project Coordinator Date Signed:

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 Week Number: 7

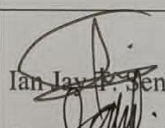
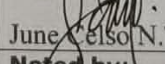
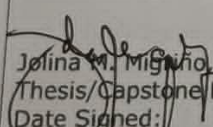
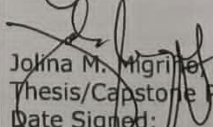
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Core Module Development (Auto-Battler AI): - Developed the asynchronous artificial intelligence scripts for the Combat Phase. - Implemented autonomous pathfinding for both Katipunan and Spanish AI units. - Programmed the base damage calculation formulas based on unit statistics and equipped weapons.	
Prepared by: Kevin Ryan N. Alabarca Francis Joyno Mareon S. Pacete	Ian Jay V. Senador June Cesar W. Tahil
Checked by:  Jolina M. Mignino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Mignino, LPT Thesis/Capstone Project Coordinator Date Signed:

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Week Number: 8

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Economy and UI Integration: - Developed the offline Random Number Generation (RNG) Gacha module for hero recruitment. - Set up the local SQLite database to track the player's Reales (currency), Rations (energy), and unit rosters. - Integrated the interactive historical pop-up quiz module that pauses combat to dispense rewards.	
Prepared by: Kevin Ryan N. Alabarca Francis Joy V. Joyno Mareo S. Pacete	 Ian Jay P. Senador  June Celso N. Tabil
Checked by:  Jolina M. Mignino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Mignino, LPT Thesis/Capstone Project Coordinator Date Signed:

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Week Number: 9

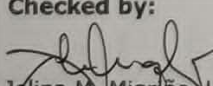
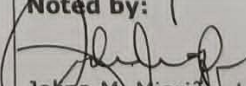
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Kapatiran Synergy Integration & Balancing: - Programmed the "Kapatiran" (Brotherhood) system, calculating proximity between deployed historical units to trigger active combat buffs. - Conducted internal playtesting to balance unit statistics and enemy wave difficulty. - Imported and finalized 2D character sprites into the 3D isometric environment.	
Prepared by: Kevin Ryan N. Alabarca Francisco V. Joyno Mareon S. Pacete	Ian Jay T. Senador June M. Tabil
Checked by: Jolina M. Migrifio, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by: Jolina M. Migrifio, LPT Thesis/Capstone Project Coordinator Date Signed:

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Week Number: 10

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Mock Defense Preparation: - Refined the capstone manuscript (Chapters 1-3) to ensure alignment with the current Unity build. - Prepared the PowerPoint presentation and system demonstration. - Conducted a Mock Defense to present the game's core functionalities to the reviewing panel and gathered initial feedback.	
Prepared by: Kevin Ryan N. Alabarca Francis John V. Joyno Mareon S. Pacete	Ian Jay B. Senador June Celso N. Tahil
Checked by:  Jolina M. Migrino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migrino, LPT Thesis/Capstone Project Coordinator Date Signed:

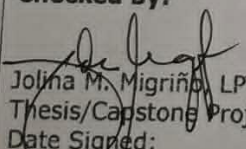
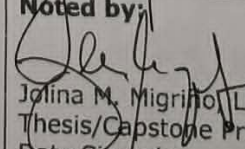
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Week Number: 11

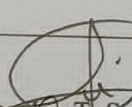
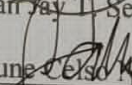
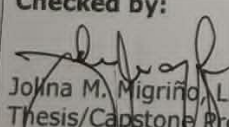
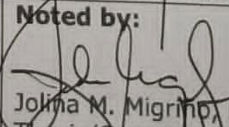
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Official Capstone Proposal Defense: - Finalized presentation materials and conducted final group rehearsals. - Officially presented "Binakayan Rising" to the Capstone Project Review Panel. - Documented all required major revisions mandated by the panel (Revision Lists 1, 2, and 3) during the defense.	
Prepared by: Kevin Ryan N. Alabarca Francis John V. Joyno Marjorie Ann S. Pacete	Ian Ray T. Senador June Celso N. Tabil
Checked by:  Jolina M. Migrino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migrino, LPT Thesis/Capstone Project Coordinator Date Signed:

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 Week Number: 12

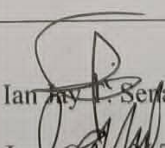
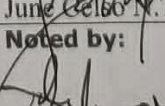
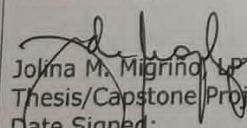
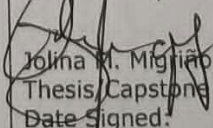
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Major Manuscript Revisions (Chapters 1 & 2): - Revised the Project Context to establish the problem starting from the international setting down to the national and local levels. - Added international and national related systems (e.g., Arkknights, Assassin's Creed, Anito) to the RRL. - Corrected the technicalities of the paper by strictly applying the APA 7th Edition format for all citations and references.	
Prepared by: Kevin Ryan N. Alabarca Francis V. Joyno Marian S. Pacete	 Ian Jay T. Senador  June Celso N. Tahil
Checked by:  Joanna M. Migrino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Joanna M. Migrino, LPT Thesis/Capstone Project Coordinator Date Signed:

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 Week Number: 13

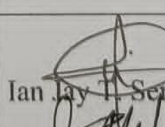
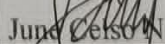
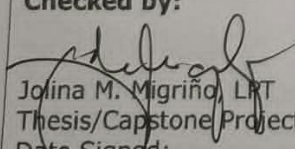
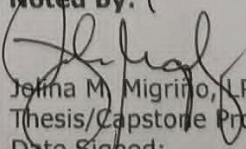
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Major Manuscript Revisions (Chapter 3): - Revised the Requirements Analysis section to explicitly list and discuss the data-gathering procedures and tools (Key Informant Interview Questionnaire). - Redesigned the Use Case Diagram to properly separate the "Player (Tactician)" and "Game AI System" actors. - Secured the required physical signatures and certificates from the historical consultant.	
Prepared by: Kevin Ryan N. Alabarca Francis John V. Joyno Marech John S. Pacete	 Ian Ray C. Serador  June Celso N. Tabil
Checked by:  Jolina M. Migriño, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migriño, LPT Thesis/Capstone Project Coordinator Date Signed:

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Week Number: 14

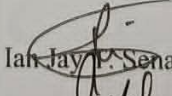
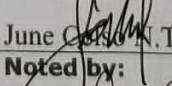
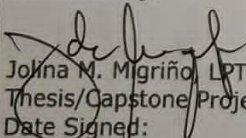
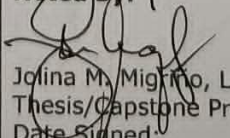
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Game Design Document (GDD) Expansion: - Restructured the GDD (Appendix F) to comply with panel feedback. - Specified the game's Lore and Theme, detailing the 4 Acts of "The Architect's Ascent." - Explicitly listed the Quests and Sub-quests for Levels 1 to 3. - Detailed the Auto-Battler Combat Mechanics and offline economy structure.	
Prepared by: Kevin Ryan N. Alabarca Francis V. Joyno Mareon S. Pacete	 Ian Jay T. Senador  June Celso N. Tahil
Checked by:  Jolina M. Migrino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migrino, LPT Thesis/Capstone Project Coordinator Date Signed:

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Week Number: 15

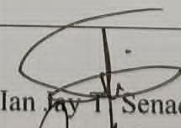
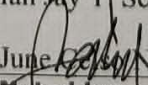
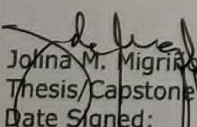
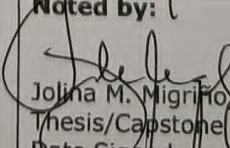
ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
System Polishing & Tables Generation: • Repolished the game's UI and fixed minor runtime errors in the Unity build. • Generated the required project tables for Appendix A, including Hardware/Software Requirements, Terrain Modifiers, Kapatiran Synergies, and the Evaluation Rubric.	
Prepared by: Kevin Ryan N. Alabarca Francis John V. Joyno Mareo John S. Pacete	 Ian Jay P. Senador  June Carlo M. Tabil
Checked by:  Jolina M. Migrino, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migrino, LPT Thesis/Capstone Project Coordinator Date Signed:

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 Week Number: 17

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Project Finalization & Submission: - Finalized the overall manuscript as a complete, single soft copy (PDF). - Conducted a final quality check to ensure all panel revisions were successfully integrated. - Submitted the final manuscript softcopy ahead of the official May 23, 2026 deadline.	
Prepared by: Kevin Ryan N. Alabarca Francis John V. Joyno Mareon S. Pacete	 Ian Jay T. Senador  June Celvin N. Tabil
Checked by:  Jolina M. Migriño, LPT Thesis/Capstone Project Adviser Date Signed:	Noted by:  Jolina M. Migriño, LPT Thesis/Capstone Project Coordinator Date Signed:

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APPENDIX L. CURRICULUM VITAE

Curriculum Vitae of
MAREON JHAN S. PACETE
NLSA ROAD, PRK 5, PINEDA COPM., Lagao G.S.C

Mjpacete05@gmail.com

09700718420/09943736515



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI General Santos City
Vocational/Technical	N/A	N/A
High School	2017-2022	General Santos City National School of Arts and Trades
Elementary	2010-2015	Jose C. Catolico Sr. Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
		month year
		month year
		month year

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

Curriculum Vitae of

Curriculum Vitae of
JUNE CELSO N. TAHIL

Purok Villegas 5th street Brgy. San Isidro

jctahil9@gmail.com

09851866828



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI General Santos City
Vocational/Technical	N/A	N/A
High School	2017-2022	Lagao National High School
Elementary	2010-2015	Dadiangas Heights Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
		month year
		month year
		month year

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
FRANCIS JOHN V. JOYNO

Mabuhay uson phase 5 prk 21 brgy mabuhay, gsc

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09916738956

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI General Santos City
Vocational/Technical	N/A	N/A
High School	2017-2022	Cronasia Foundation College
Elementary	2010-2015	Notre Dame Sienna College, GSC

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

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AFFILIATIONS

Inclusive Dates	Name of Organization	Position
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month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

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		month year
		month year
		month year

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Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first)

Curriculum Vitae of
KEVIN RYAN N. ALABARCA

Prk.2d Homeless, Katangawan G.S.C

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09611026290



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2025-2027	STI General Santos City
Vocational/Technical	N/A	N/A
High School	2019	Johnny Ang National High School
Elementary	2015	Datu Molod Integrated School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
month year		
month year		
month year		
month year		

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		month year
		month year
		month year

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Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
IAN JAY T. SENADOR

Foremost Village Phase 1, Apopong G.S.C

Ianjaysenador4638@gmail.com

09093451163



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2024-2027	STI General Santos City
Vocational/Technical	N/A	N/A
High School	2017-2022	HolY Trinity College of General Santos City
Elementary	2016-2017	Dadiangas Heights Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

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		month year
		month year

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month year	
month year	
month year	

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